

Theopractical Sheet 1

1. A tool of which *analysis type* of digital forensics should you choose to investigate the contents of RAM modules?

Ans:Live Forensics

We need to examine volatile data, which vanishes when the system is turned off in order to look into the contents of RAM modules because it deals with analyzing a system while it is still operating and gives access to RAM contents like open files, running processes, network connections and encryption keys hence live forensics is the appropriate analysis type.

2.You are comparing a CD-ROM containing a video game and the contents of the CPU flags register. In which of the classifications of digital traces (that we discussed in the lecture) will the two objects end up in different classes? Assume that any necessary information for this question is given, hence exclude classifications that would require more details.

Give your answer as a JSON array of strings. Refer to the naming on the slides. Your answer(s) will be matched using regular expressions. Give a short justification in your proof-of-work file.

["Classification by A", "Classification by B", "Classification by C"]

Ans:["Classification by Volatility"]

From the slides the data can be catagorised as Persistent Volatile(CD-ROM) and extremely volatile(CPU Flags) respectively.Both are catagorised under volatile.

The degree of volatility

CD-ROM= Persistent (data remains without power) CD-ROM

CPU flags register = extremely volatile (data is immediately lost when the power is cut off).

3.Recall the CSI model.

a) What do C, S and I stand for? (1 point)

Give your answer in the following format and pay attention to the whitespaces and dashes. This exercise is binary graded.

Classy - Sunny - Island

Ans:Claim - Support - Information

From the slides I understood that Abstract model of traces or evidence

Each trace consists of three parts: C, S, I

- **Claim:** Metadata
- Description of the provenance
- **Support:** Matter
- Data storage
- **Information:** Interpretation of the data
- Meaning of the data

b)What do we express with I? (1 point) There are more than one boxes to tick here.

Ans:The meaning of the data

From the above question we understood that I stands for information which stands for the meaning of the data.

4. Assume the following:

The probability that a randomly selected IP address has won in a web lottery is 0.05.

The probability that a randomly selected IP address has won in the lottery and is part of a botnet is 0.01.

The probability that a randomly selected IP address is part of a botnet is 0.2.

a) What is the probability that a winning IP address is part of a botnet? Give the result as a decimal number. Use a dot "." as a decimal point. (1 point)

Ans:We use Bayes' Theorem: $\Pr(\text{Botnet} \mid \text{Win}) = \Pr(\text{Botnet} \cap \text{Win}) / \Pr(\text{Win})$

From the given data $\Pr(\text{Botnet} \cap \text{Win}) = 0.01$

$\Pr(\text{Win}) = 0.05$

We get:- $0.01/0.05 = 0.2$

b) What is the probability that an IP address that is part of a botnet has won in the lottery? Give the result as a decimal number. Use a dot "." as a decimal point. (1 point)

Ans: From the values given we can get

$\Pr(\text{Win} \mid \text{Botnet}) = \Pr(\text{Botnet} \cap \text{Win}) / \Pr(\text{Botnet})$

$= 0.01/0.2$

$= 0.05$

c) To what extent should probabilities matter in jurisdiction? (Up to 2 bonus points, manually graded)

Ans: From the Sally Clark example which was discussed in the class I understood that probabilities can be used to argue for or against evidence they shouldn't be the only factor to consider in court cases. Courts must also take other important factors like fairness, context, and intent into account. Even though a high probability might suggest guilt it conclusively does not prove it particularly if the pattern also applies to innocent people. Hence I feel Probabilities can be used as an addition to an argument but not as a sole factor in court cases. It should be used carefully with other evidences.

5. According to the classification by volatility, to which class belong EEPROM modules? This exercise is binary graded. There is one correct option.

Ans: Persistent as it retains data when power is turned off.

6. SSD controllers pretend to be control and HDD.

7. Assume the following:

Sectors have a size of 512 bytes each.

A cluster consists of 8 sectors. $8 \times 512 = 4096$ bytes

Our file has a size of 777 bytes.

This exercise is binary graded.

a) How large is the RAM slack in bytes?

Ans:RAM slack is usually the unused space in the last sector that contains the file.

Given our file size is 777 bytes and each sector can hold 512 bytes so the first 512 bytes will go into sector 1 and the remaining $777-512=265$ bytes will go into sector 2. But as we know there is still some unused space left in sector 2 this will be occupied by the RAM slack which will be:

$$512-265=247 \text{ bytes}$$

b) How large is the Drive Slack in bytes?

Ans:Drive Slack is the space that is completely unused in the remaining cluster. So we know the file occupies 2 sectors from the above question a) and we also know that 1 cluster contains 8 sectors so there are 6 empty sectors which will be used for Drive Slack that is:

$$6 \times 512 = 3072 \text{ bytes}$$

c) How large is the File Slack in bytes?

Ans:File Slack is nothing but RAM slack+ Drive Slack

$$\text{File Slack} = \text{RAM Slack} + \text{Drive Slack}$$

As we know the values of RAM Slack and Drive Slack from a) and b) we put the values we get:

$$= 247 + 3072$$

$$= 3319 \text{ bytes}$$